

1. System Architecture Overview

GOLDENERA is an account-based blockchain utilizing a state-machine architecture. It does not support programmable smart contracts. Instead, it features native support for custom tokens and a built-in governance system (BIP - Blockchain Improvement Proposals) directly within the protocol layer.

- **Type:** Account-based.
- **State Storage:** Merkle Patricia Trie.
- **Smart Contracts:** None.
- **Native Features:** Embedded Custom Tokens, On-chain Governance (BIP).

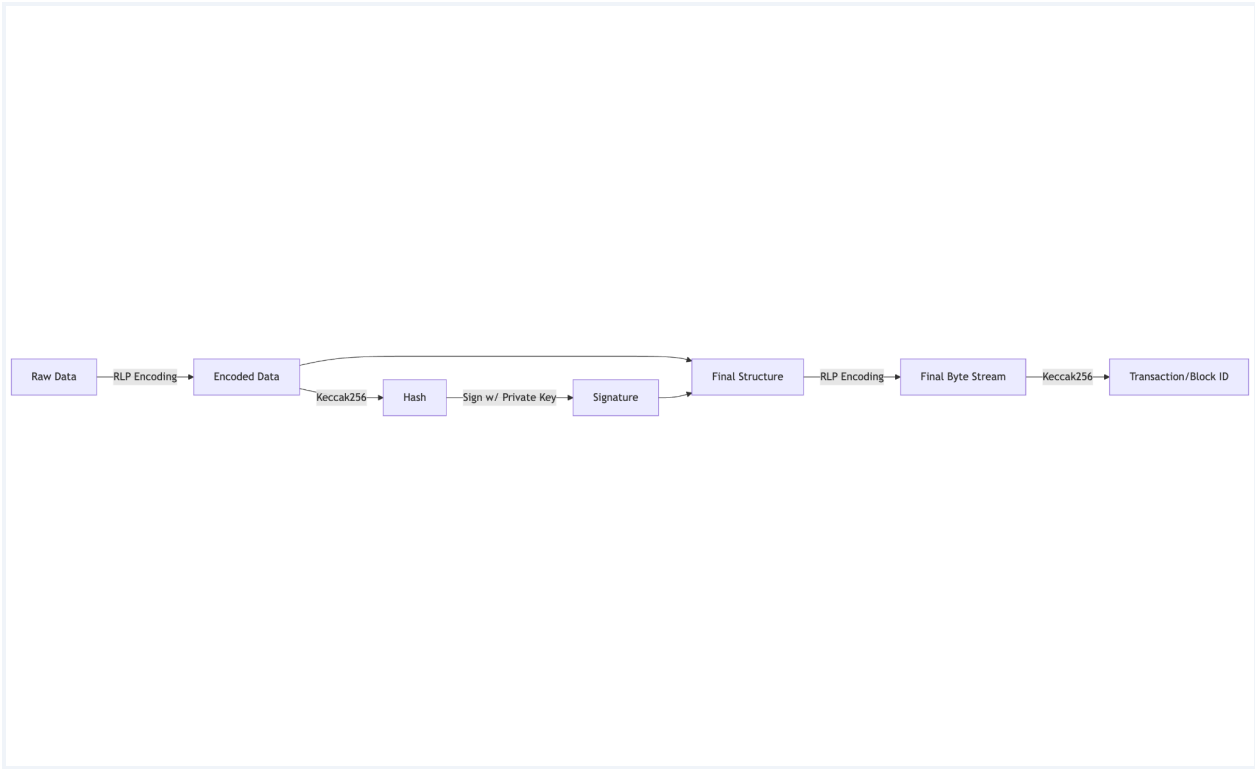
2. Cryptography and Encoding

- **Hashing Algorithm:** Keccak256 (used for block hashes, transaction hashes, and state keys).
- **Signature Scheme:** ECDSA (Ethereum-style) using the secp256k1 curve.
- **Data Encoding:** RLP (Recursive Length Prefix).
- **Addressing:** Ethereum-style address derivation (20-byte).

Hashing Process (ID Generation)

The ID of a block or transaction is derived as follows:

1. **Raw Data** (excluding signature) is RLP encoded.
2. **Hash** is calculated using Keccak256.
3. **Signature** is applied using the Private Key.
4. **Final Object** (Raw Data + Signature) is RLP encoded.
5. **Final ID** is the hash of the signed RLP object.



3. Global State Structure

The global state is stored in a Key-Value store structure (Merkle Patricia Trie). The keys are specific to the data type being stored.

State Mapping Definitions

State Entity	Key Definition	Value / Content
Network Params	Global Constant Key	Consensus rules (Reward, Fees, Difficulty params).
Authority	AuthorityAddress	Boolean/Flag (is authorized).
Validator	MinerAddress	Boolean/Flag (is allowed to mine).
BIP State	TransactionHash (of BIP_CREATE)	Status (APPROVED/DIS/PENDING),

		Votes, Expiration.
Token State	Derived Address*	Token Properties (Name, Supply, URLs, etc.).
Address Alias	AliasString (e.g., "blockRewardPool")	The corresponding Wallet Address.
Account Balance	Hash(Address + TokenAddress)	Balance amount (Wei).
Account Nonce	Address	Transaction counter (BigInteger).

*Token Address Derivation: Calculated deterministically using Keccak256(SenderAddress + TransactionNonce) from the BIP_CREATE transaction.

4. Consensus and Mining

Algorithm

- **Proof of Work:** RandomX (ASIC-resistant).
- **Difficulty Adjustment:** ASERT (Absolute Sahptouris Exponential Recalculation Target).
 - Adjusts difficulty every block.
 - Based on TargetMiningTime defined in Network Params.

Permissioned Mining

Mining rights are restricted based on the **Validator State**.

- **Standard Operation:** Only addresses present in the Validator List can mine blocks.
- **Fallback Operation:** If the Validator List is **empty**, mining is public (anyone can mine).

Block Rewards

Rewards are dispensed based on NetworkParams.

- **Source:** BlockRewardPoolAddress.
- **Mechanism:**
 - If BlockRewardPoolAddress is 0x000...: Tokens are minted infinitely (inflationary).
 - If BlockRewardPoolAddress is a specific address: Reward is deducted from this address's balance.
 - If the pool address is empty: Miner receives only transaction fees.

5. Governance: BIP System (Blockchain Improvement Proposals)

The blockchain creates and executes changes via specialized transactions. Only **Authorities** can create BIPs and vote on them.

- **Role:** Authority (Address + Private Key).
- **Mechanism:** On-chain voting.
- **Threshold:** >51% (Simple Majority) of APPROVAL or DISAPPROVAL to finalize a state change.

BIP Lifecycle

1. **Creation:** Authority sends BIP_CREATE transaction.
2. **Voting:** Authorities send BIP_VOTE transactions referencing the BIP_CREATE hash.
3. **Execution:** Automatic upon reaching threshold or expiration.

BIP Types (Operations)

ID	Type	Description
0	AUTHORITY_ADD	Adds a new Authority address.
1	AUTHORITY_REMOVE	Removes an existing Authority.
2	ADDRESS_ALIAS_ADD	Maps a string alias to an address (DNS-like).
3	ADDRESS_ALIAS_REMOVE	Removes an alias.
4	TOKEN_CREATE	Initializes a new custom token.
5	TOKEN_UPDATE	Updates mutable token properties (URLs).
6	TOKEN_MINT	Mints new tokens (if mutable).

7	TOKEN_BURN	Burns tokens (reduces supply).
8	NETWORK_PARAMS_SET	Updates consensus parameters (fees, rewards, etc.).
9	VALIDATOR_ADD	Adds a miner to the allowlist.
10	VALIDATOR_REMOVE	Removes a miner from the allowlist.

6. Token Standards

The blockchain supports a native token (Address 0x00...00) and custom tokens.

- **Native Token:** Used for all transaction fees.
- **Custom Tokens:** Created via TOKEN_CREATE BIP.
- **Token ID:** 20-byte Address.

Token Properties

- Name (String, Updatable via BIP)
- SmallestUnitName (String, Updatable via BIP)
- NumberOfDecimals (Int, Immutable)
- WebsiteUrl (String, Updatable via BIP)
- LogoUrl (String, Updatable via BIP)
- MaxSupply (BigInteger, Immutable)
- IsUserBurnable (Boolean, Immutable) - If true, users can send to 0x00 to burn.
- TotalSupply (Wei, Updatable only via Mint/Burn BIPs or Mining logic).

7. Transactions

Structure

- **Type:** TRANSFER, BIP_CREATE, or BIP_VOTE.
- **Nonce:** Strictly incremental (PreviousNonce + 1).
- **TokenAddress:** Must be specified for every transaction (even native transfers use 0x00...00).
- **Fee Model:** Paid by Sender.
 - $TotalFee = BaseFee + (ByteSize * ByteFee)$.

- Fees are always paid in the **Native Token**.

Mempool & Node

- **Mempool:** Standard propagation of valid, unconfirmed transactions.
- **Nodes:** Open source. Any node can validate and propagate. Only Validators can mine (produce blocks).

8. Network Parameters (Dynamic)

These values are stored in the state and can be modified at runtime via NETWORK_PARAMS_SET BIP.

- **Block Reward** (Wei): Defines the fixed amount of native currency awarded to a miner for successfully mining a valid block.
- **Block Reward Pool Address** (Address): Determines the source of the block reward.
 - If set to 0x00...00 (Null Address): The reward is newly minted (inflationary model).
 - If set to a specific address: The reward is deducted from this address's balance. If this address has insufficient funds, the miner receives only transaction fees.
- **Target Mining Time Ms** (long): The desired time interval (in milliseconds) between the generation of two blocks. This value is used by the difficulty adjustment algorithm to regulate network hashrate.
- **ASERT Half Life Blocks** (long): A parameter for the ASERT difficulty algorithm that acts as a smoothing factor or "relaxation time." It determines how responsive the difficulty adjustment is to recent fluctuations in hashrate.
- **ASERT Anchor Height** (long): A specific block height used as a fixed reference point (checkpoint) for the ASERT algorithm to calculate the next required difficulty.
- **Min Difficulty** (BigInteger): The absolute minimum difficulty value allowed by the network. The difficulty adjustment algorithm is prevented from setting a target below this threshold.
- **Min Tx Base Fee** (Wei): The mandatory flat fee charged for every transaction processed by the network, regardless of the transaction's size or type.
- **Min Tx Byte Fee** (Wei): The variable fee charged per byte of data in a transaction. The total variable cost is calculated as $\text{TransactionSizeBytes} * \text{MinTxByteFee}$.